

LearnThatDance: Augmenting TikTok Dance Challenge Videos with an Interactive Practice Support System Powered by Automatically Generated Lesson Plans

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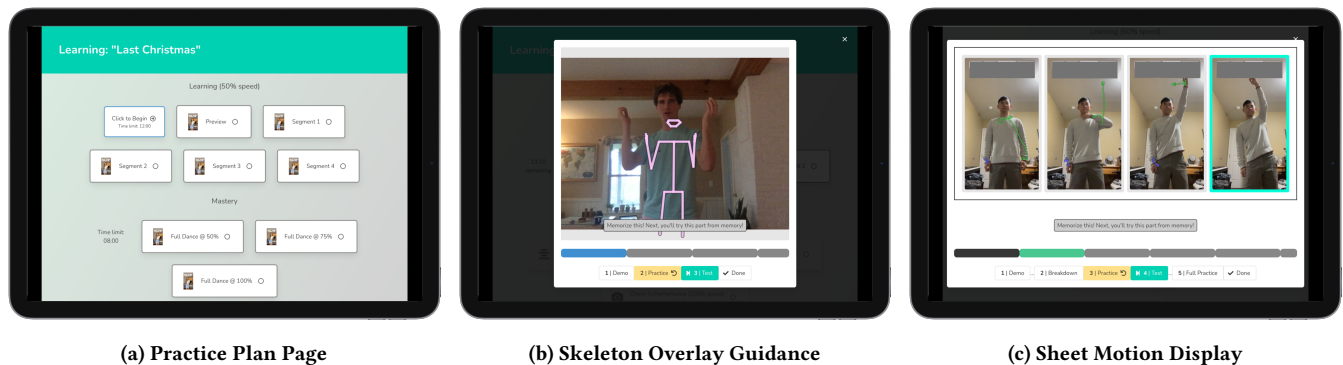
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(a) Practice Plan Page

(b) Skeleton Overlay Guidance

(c) Sheet Motion Display

Figure 1: Dance Learning Web App

ABSTRACT

In this demo we showcase an interactive application to support the learning of “TikTok dance challenge” short dance choreographies. Our system utilizes dance challenge videos as the information source, performing music analysis and pose estimation to segment the dance into learnable chunks and generate a practice plan that implements motor learning techniques such as incremental part-learning and fading guidance. These plans are presented in a web app that implements video demonstration, augmented webcam mirroring, practice recording/review functionality, and both concurrent and terminal feedback. By operating on a ubiquitous information

source, generating the lessons automatically, and requiring only a web browser and webcam in the user interface, our system is a step towards significantly expanding the reach of dance choreography learning and a platform for further research into dance HCI.

KEYWORDS

dance, dance education, intelligent tutoring system

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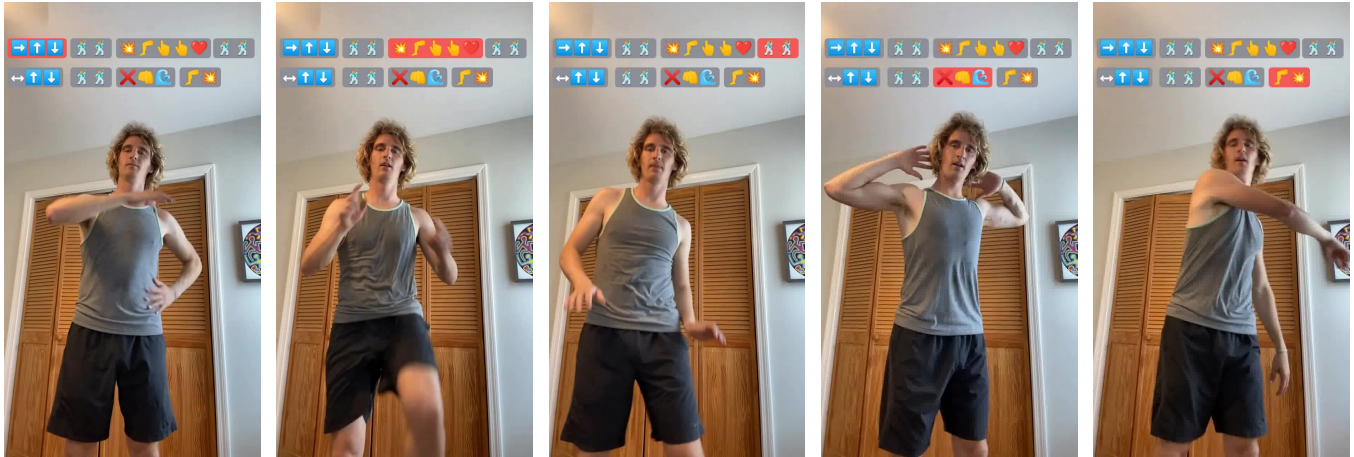


Figure 2: TikTok Dance Tutorial Video Format. The videos are overlaid with boxes representing a segmentation of the dance, which contain emoji labels of the dance moves. The boxes change background color in as the dance progresses.

1 INTRODUCTION

With over 53.8 billion views [4], *#dancechallenge* is one of the most popular trends on TikTok, whereby users come up with short choreographed routines and challenge each other to learn them. On the TikTok platform, some dance challenges are formatted as tutorials with rectangles overlaid on the video containing emoji and text labels of dance moves, which are time-synced to the video - changing background color when the corresponding dance move is being performed (Fig. 2). These tutorial videos are popular, with over 18.7 billion views for the *#dancetutorial* tag [5]. Still, these tutorial videos have limitations as they do not provide guidance to support people as they learn dance challenges, and only a fraction of shared choreographies have tutorial versions available.

Learning dances has been shown to have benefits for people across the age spectrum – from assisting socio-emotional learning in school children [9] to improving quality of life in patients with Parkinson’s disease [15]. Given these benefits, we see an opportunity to apply a grounding in motor learning science and insights from prior dance HCI work to create an enhanced learning experience for TikTok dance challenges. With recent progress in lightweight pose-estimation techniques, we can implement this experience on devices that many people already own, such as tablets and laptops, and can support the learning of dance choreographies without requiring manual tutorial creation.

2 RELATED WORK

Researchers have developed numerous tools to aid with dance performance, composition, and analysis [10], but relatively few for helping teach dance skills to beginners. The ones that have been developed typically require specialized hardware [1, 8, 13] or hand-crafted content [6, 17], limiting their reach. There are some commercial digital products that provide a dance learning experience. These generally fall into one of two categories: games designed for entertainment, such as Just Dance and Dance Dance Revolution, and products designed for dance learning, such as STEEZY Studio and learntodance.com. These products operate on manually-created

content, adding to their expense and restricting their ability to scale to the expansive array of dance styles and skills that exist.

3 SYSTEM DETAILS

Our system is implemented as two components. The first component is a data processing pipeline that accepts video files containing dances and produces practice plans for teaching them. The second component is a learning interface implemented as a web application that guides learners through the pre-computed practice plans. To ground our system design, we drew from learnings in motor learning literature and applied them to our practice plan, as shown in Fig. 3.

- *Incremental part learning* - In a meta-analysis, Fontana et al. [3] found evidence that part learning - breaking down a

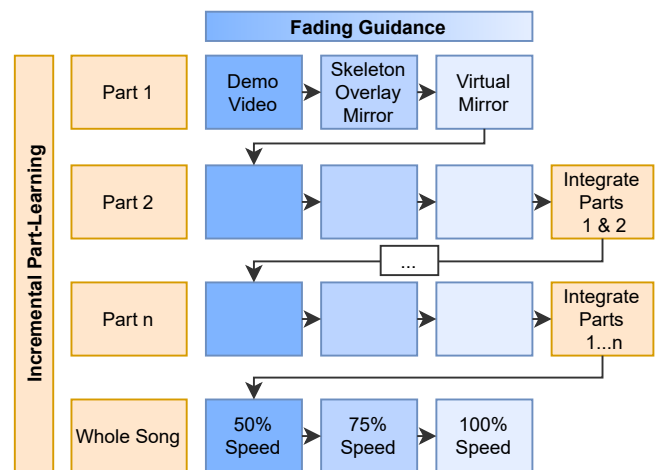


Figure 3: Practice Plan Schematic. The practice structure incorporates two techniques drawn from motor learning literature: fading guidance and incremental part-learning.

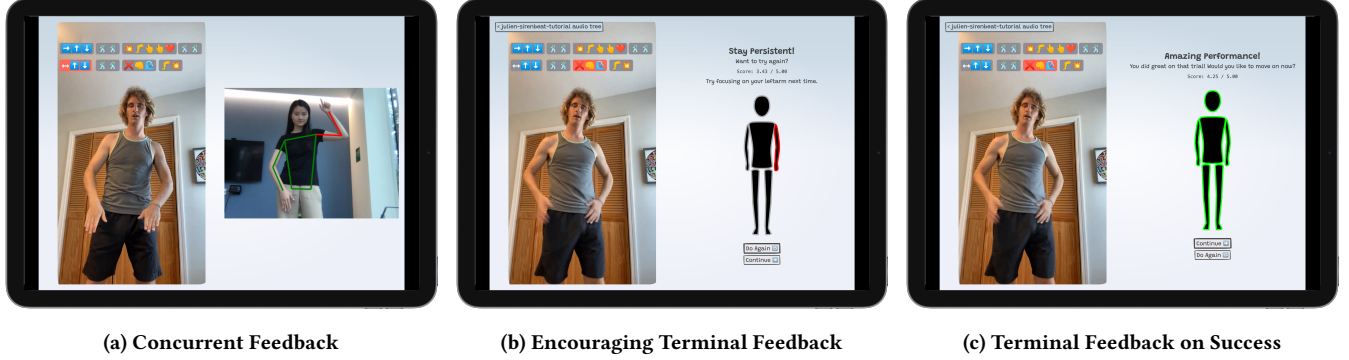


Figure 4: Feedback. Both concurrent and terminal feedback are offered.

task into pieces and training them separately - was superior to whole learning for tasks such as dance (this is also the method that professional dancers use to learn choreographies [11]). Drawing from this, we adopt a version of part learning that incrementally teaches new segments of the dance.

- *Fading guidance* - In a their review, Sigrist et al. [16] report that concurrent feedback can be helpful in the initial stages of learning a complex task, but can be counterproductive later on, in line with the *guidance hypothesis* [12]. Therefore, we adopt a practice plan sequence that progressively reduces the level of task support as the dance is learned.

The lesson compilation pipeline consists of multiple stages of data processing where the last stage produces practice plans that are later consumed by the learning interface (see Fig. 3). The first step is to extract a motion representation from the video. We use the MediaPipe framework [7] to extract a skeletal pose of the dancer in each frame. In the second stage, we analyze multiple aspects of the dance with the objective of producing a higher-level, semantically meaningful understanding of the motion. We compute: (a) the speed minima of a subset of joints in 3D space, (b) tempo of the accompanying music, and (c) simplified motion trails of the hands. The decomposition algorithm then chooses a decomposition strategy for the dance based on these features, preferring to use a tempo-based segmentation of 4-beat measures but falling back to a speed-minima segmentation in the case of an undetectable tempo.

In the third (and final) stage, a sequence of practice activities is assembled based on this segmentation. The assembler first creates a learning phase, designed to introduce a dance to a user who has never seen it before. The first activity is a preview of the dance, followed by a sequence of activities that incrementally train the dance segment-by-segment, following an *incremental part learning* approach. For each of these activities, the user sees: (1) a video demonstration of the segment, (2) a practice stage with either a skeletal overlay or sheet motion learning aids (described below), (3) a test stage, in which the user sees their webcam feed and performs the segment from memory, (4) an integration stage, in which the user performs the entire dance up until the end of the newly learned segment. All of these activities in the learning phase occur at half

speed. By the end of the learning phase, users have been introduced to the entire dance in pieces and have performed integrative steps to assemble the pieces. The lesson compiler then adds a mastery stage, consisting of a series of activities that prompt the learner to practice the entire dance at increasing speeds (0.5x, 0.75x, and 1x). Each of these mastery activities follows the same demo-practice-perform sequence as in the learning phase. Lessons are stored in a JSON format for use in the learning interface.

The second component of our system is the learning interface, a web application that presents and guides the user through the automatically generated lessons. Upon visiting the site, users are greeted with a lesson menu screen (see Fig. 1a) that both visually illustrates the learning path ahead of them and allows the user to navigate the learning process. Fig. 1b and Fig. 1c show the mini-lesson interface. The user is presented with a sequence of steps within the lesson and can play, repeat, and navigate between the steps. Above the buttons is a progress bar separated and color coded to the decomposition of the motion. Content is displayed at the top of the screen. This can be a demonstration video, an overlay of a skeleton from the target dance on top of the user’s webcam feed (Fig. 1b, inspired by [1, 8, 14]), or a ‘sheet motion’ display of key frames overlaid with motion trails of the hands (Fig. 1c). Two types of feedback are provided to the user: concurrent feedback, in the form of a color-coded skeleton overlay (Fig. 4a); and terminal feedback, after each dance attempt (Fig. 4b, Fig. 4c).

4 EVALUATION

We conducted a user study with $n=54$ participants (36 female, mean age 18.67 years) from our university, who used our system from home on their own devices. Participants learned three different TikTok dances, one in each of the following conditions: (1) an experimental condition using our automatically generated lessons and the skeleton overlay visual aid, (2) an experimental condition using our automatically generated lessons and the sheet motion visual aid, and (3) a control condition using a TikTok tutorial video created by a creator on the platform, with play/pause/seek controls enabled. In all conditions users were given 12 minutes to learn the dance at half speed followed by an additional 8 minutes to learn it at full speed before recording their final dance performance. We

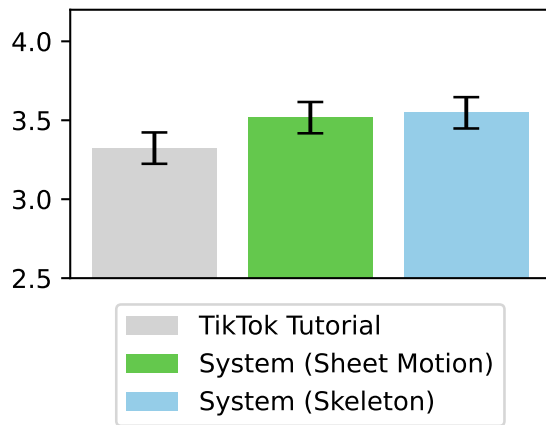


Figure 5: Learning Outcomes by Condition. Compares learning from TikTok tutorial videos to learning from our dance challenge practice system.

hired three human raters to score the dance performances based on their similarity to the TikTok tutorial video.

We fitted a linear mixed model to predict this human rated score with the fixed effect of condition and dance and the random intercepts for the participants. We then tested the fixed effects of our model with Type 3 Analysis of Variance (ANOVA) and found a significant main effect of condition, $\chi^2(2, N = 130) = 7.922, p = .019$. As seen in Fig. 5, users performed better with both versions of our practice support system than when learning from TikTok tutorial videos.

5 CONCLUSION

We have presented a system for dance choreography learning that creates practice plans from dance videos and teaches them in a web interface. This builds on the state of the art by removing barriers to adoption such as complex hardware requirements and time-consuming manual content creation, and extends our previous demo [2] with the addition of explicit feedback. Initial results indicate that participants learn more effectively from our system than from TikTok dance video tutorials. We take this as evidence that an automatic dance practice support system, guided by motor learning theory, can expand access to short choreography learning by making dance challenges easier to learn - and especially so for those challenges where a human-created tutorial video is unavailable. Ongoing work builds upon this system with new forms of feedback, tree-like dance decompositions, and a robot demonstration. Looking ahead, we see our work contributing to systems literature on dance HCI, as well as serving as a technology probe to investigate motor learning questions.

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